

Compression-torsion Coupling in the Compression Response of Tubes with Kresling Units

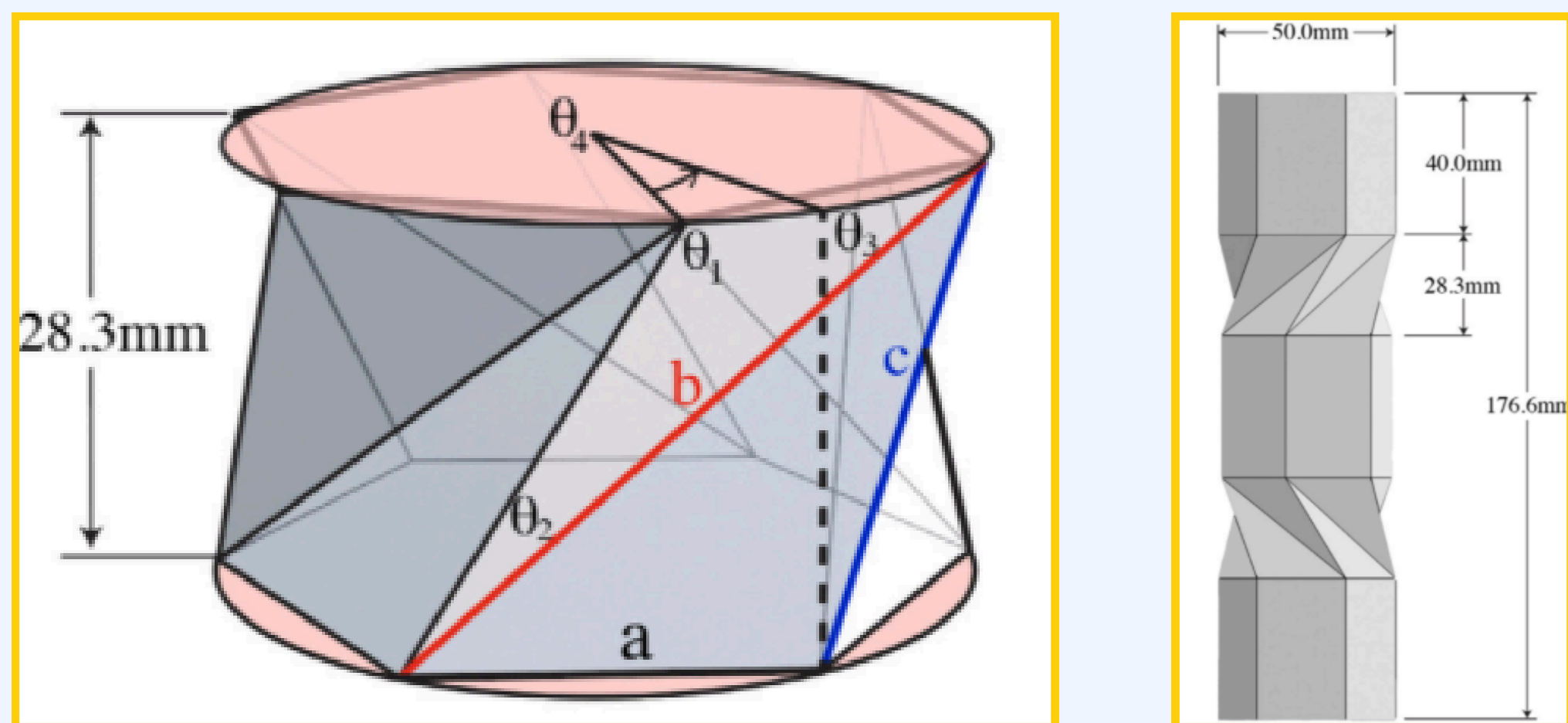
Georgina Sanchez,¹ Mio Crilley,¹ Colin Hunter,² Royan D'mello³

¹College of Engineering, University of Michigan; ²Stamps School of Art & Design, University of Michigan;

³Department of Aerospace Engineering, University of Michigan

Project Background

- Tubular structures with origami-based structural elements have unique properties and designing these can lead to tailored energy absorption under compression loading.
- Additionally, these structures exhibit coupling of compression and torsion (twisting) under axial compression loading. This research is on an origami-based structure called Kresling unit that allows for cross-sectional rotations under axial compression.
- Our goal is to find the geometric parameters to maximize energy absorption and characterize the strength and rotation of the Kresling through experimental tests on 3D printed specimens and support a simulation-based design study on 3D Kresling models.

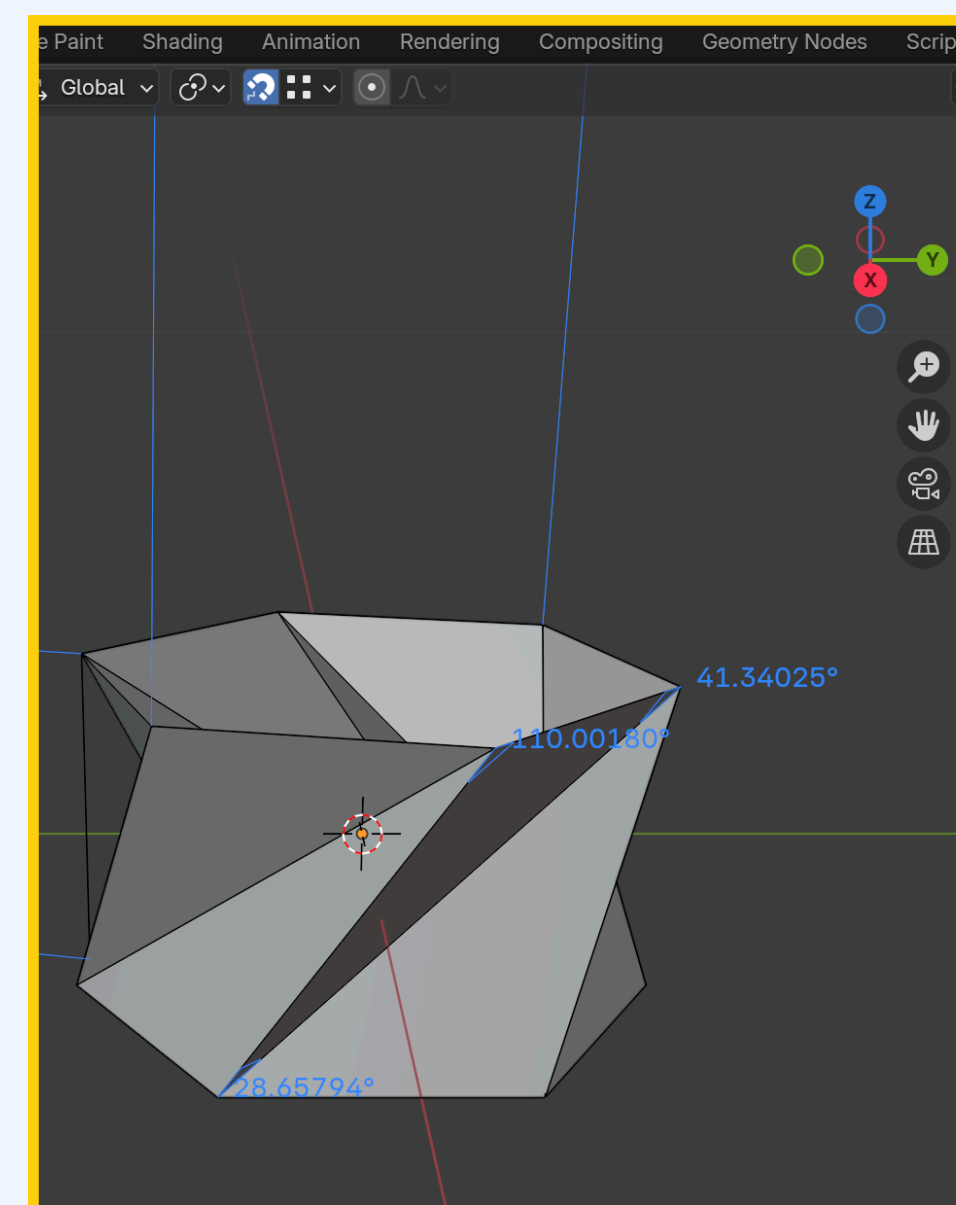


(LEFT) The dimensions of the Kresling unit studied by D'Mello and Hunter. This geometry was varied to attempt to find an elite Kresling model.
(RIGHT) A multi-Kresling stack.

- Background on Digital Image Correlations (DIC): an optical analysis method to measure surface deformation and strains by comparing snapshots of deformation in a structure under load.
- We manually selected tracking points are on the specimen's speckled surface, and recorded their displacements in both X and Y directions to monitor axial deformation, lateral expansion and twisting.

Methods

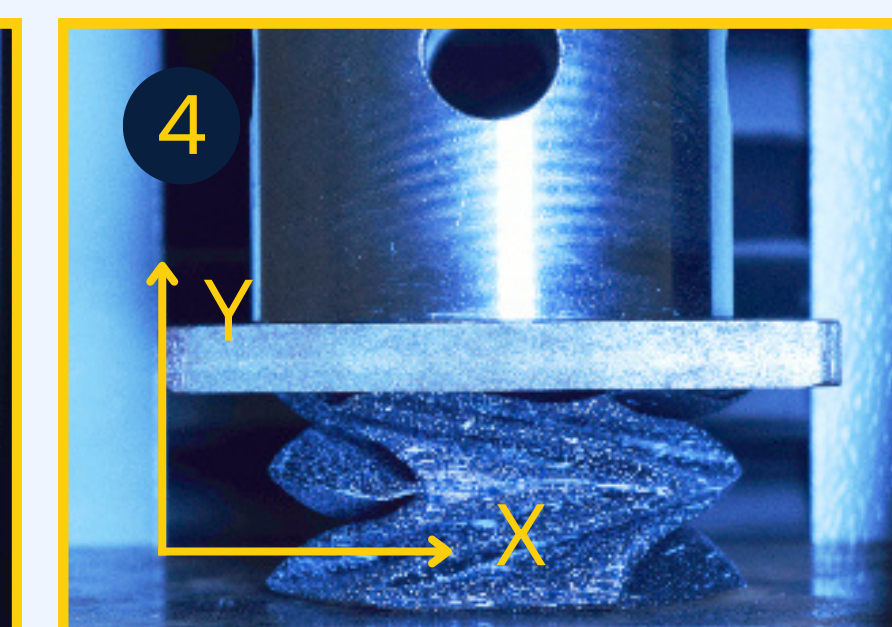
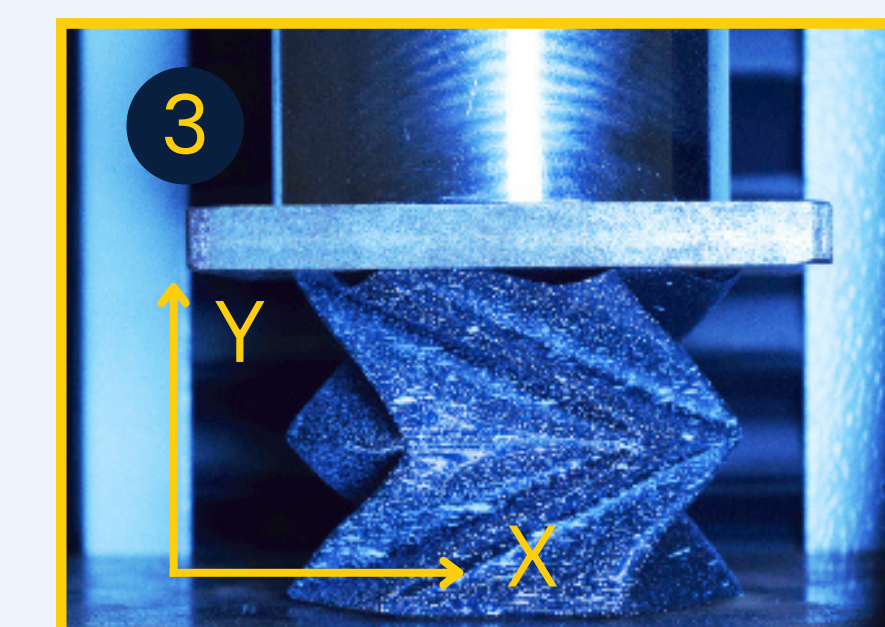
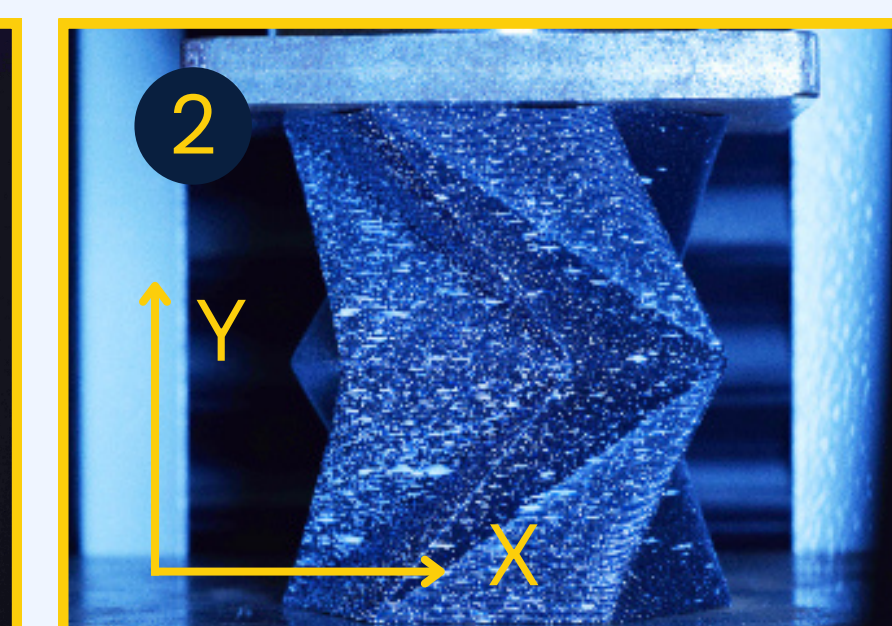
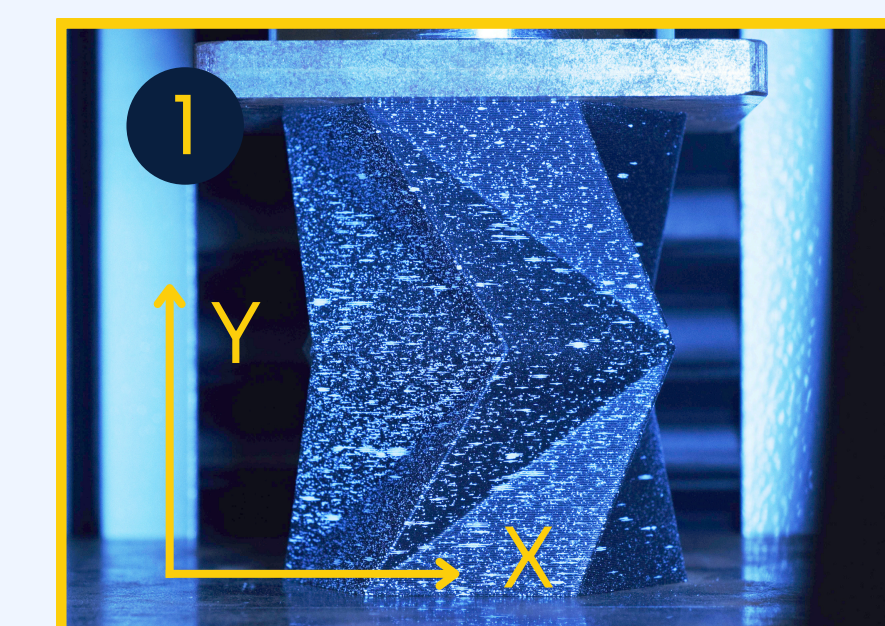
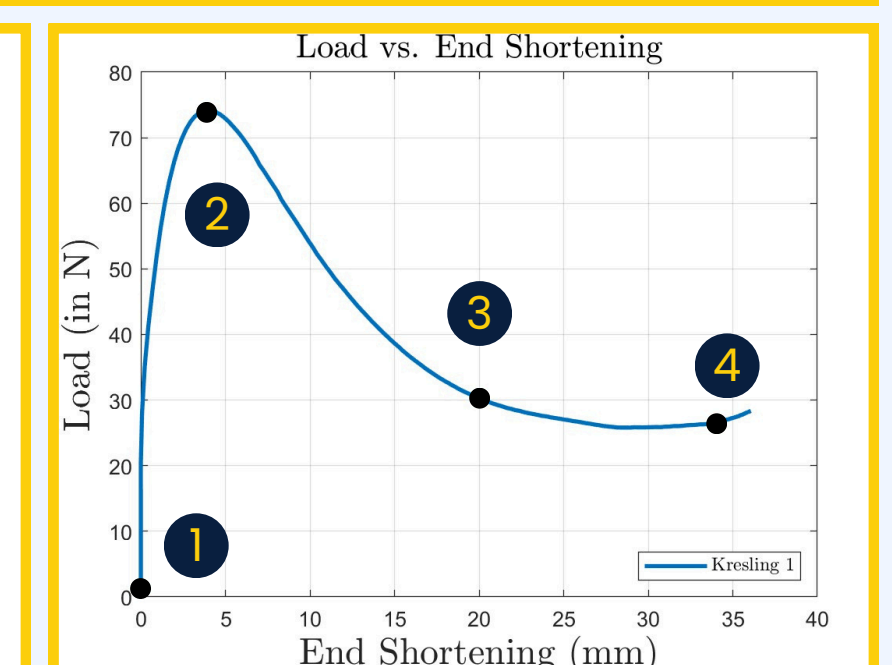
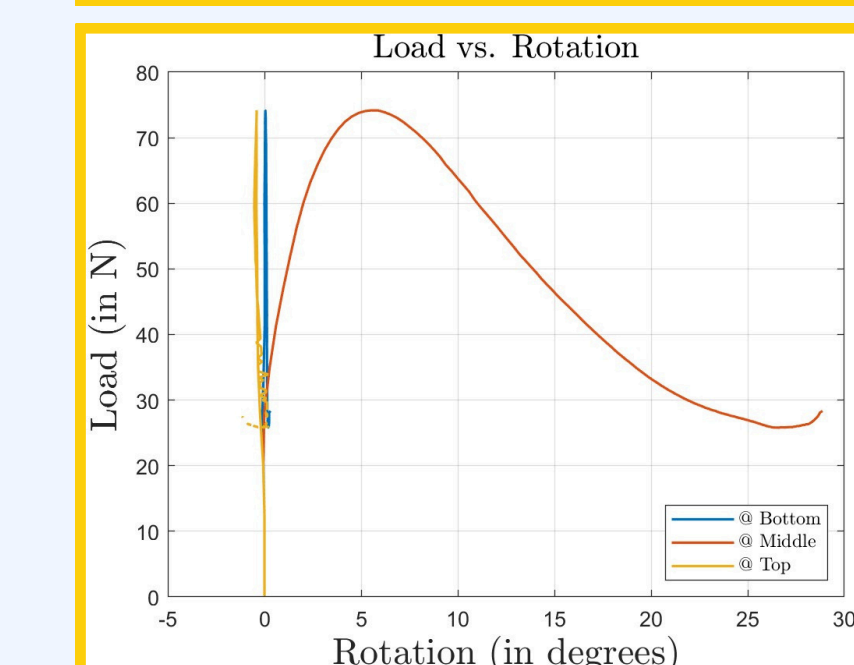
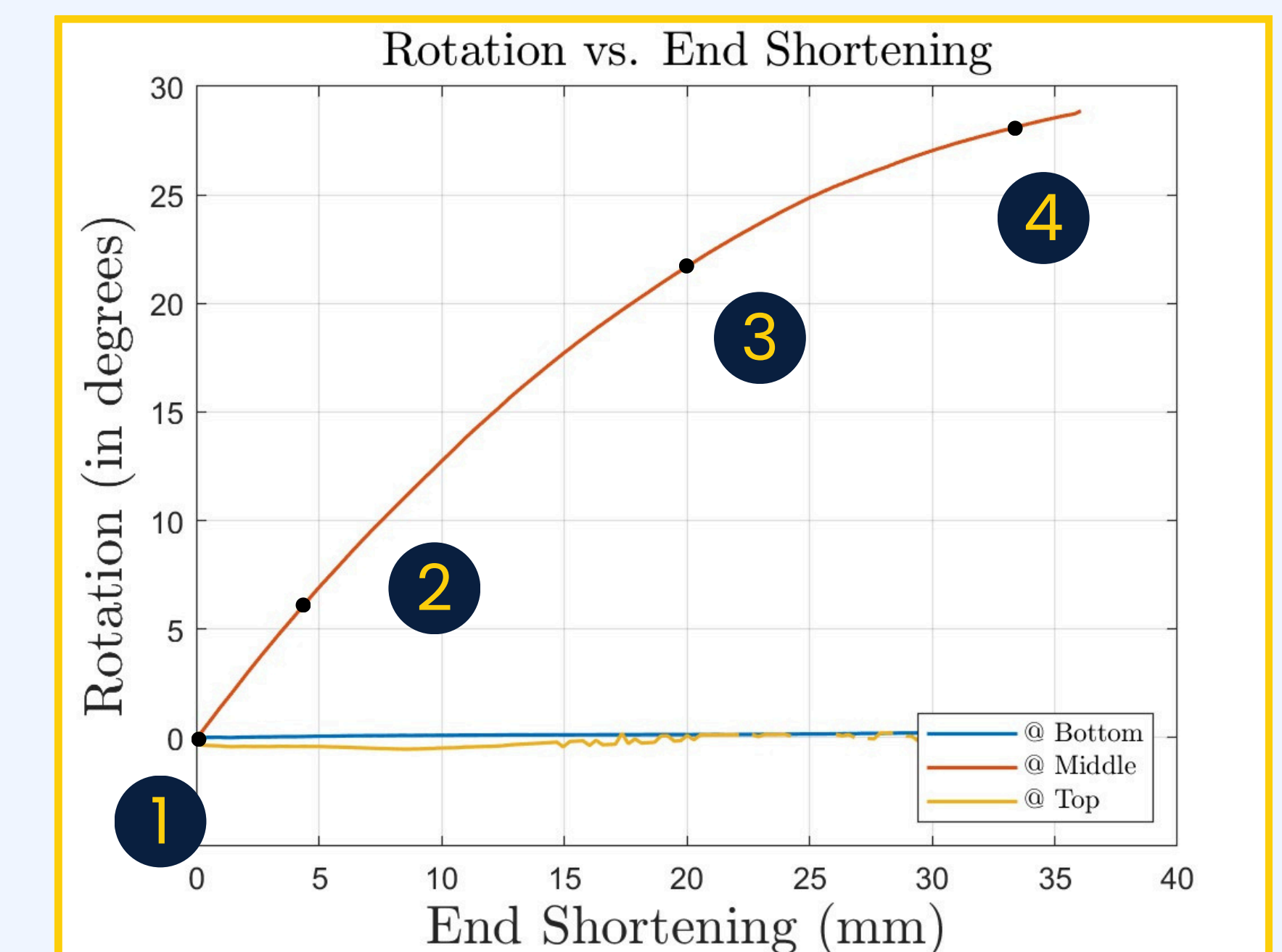
- **Blender:** a 3D modeling software to create different models of the Kresling with different interior angle measurements and scalings.
- **AutoCad:** a program to convert file formats from .stl to .iges to support virtual experiments.
- **3D Printer and load frames** (seen below): tools used to measure how much force the models could handle until failure.
- **GOM Correlate:** a program to perform DIC to analyze deformation and rotation during the experiment.
- **Abaqus:** a finite element software to conduct virtual experiments on Kresling units to analyze their behavior at different wall thicknesses, materials systems, angles, and height/width variations.



(LEFT) Screenshot taken from a Kresling unit modeled in Blender.
(CENTER) Image of the Bambu Labs X1C Printer we used to print these Kresling samples in vase mode (1 wall layer, 0.4mm nozzle, 0.2mm layer height).
(RIGHT) Picture of a Kresling unit in the load frame taken during an experiment.

Results and Conclusions

- When a single Kresling unit is axially compressed, the load response shows a peak followed by a drop along with cross-sectional rotation.
- We observe that as wall thickness of Kresling unit increases, maximum reaction force increases; and as height of Kresling unit increases, the maximum rotation increases.
- We expect to find improvements in energy absorption from the results of this research. Our learning and analysis of the 27 CAD models we designed throughout the year with varying heights and interior angle measurements, will support this research in finding the optimal design.



Response plots analyzing several features of one experiment including rotation, load, and end shortening. Points highlighted correspond to selected images taken during the experiment.

Future Work

- We will perform compression tests and DIC to compare the behavior of single Kresling units with FE simulations.
- We will analyze experimentally how stacked Kresling units with variations in design improves energy absorption and rotation response and compare with FE analysis.
- This study will serve as a guide to analyze tunable tubes with other origami structures.